SATYA LINUX NOTES

2. Linux vs Windows

2.1 Introduction

Linux and Windows are two widely used **operating systems**, each designed with different goals:

- **Linux** → Open-source, performance-oriented
- **Windows** → Commercial, user-friendly

Choice depends on **use case and requirements**

2.2 Basic Difference

Feature	Linux	Windows
Type	Open-source	Proprietary
Cost	Free	Paid license
Interface	Command-line + GUI	Mostly GUI
Usage	Servers, developers	Personal & business users

2.3 Stability

- **Linux**:
  - Highly stable
  - Rarely crashes
  - No frequent reboot required
- **Windows**:
  - Less stable under heavy load
  - Requires frequent updates/restarts

Linux is preferred for **servers**

2.4 Performance

- **Linux**:
  - Better under **high workload**
  - Efficient memory usage
- **Windows**:

    - Performs well for **general usage**
    - Slower under heavy load

Linux is used in **high-performance systems**

2.5 Scalability

- **Linux**:
    - Easily scalable
    - Supports large servers and networks
- **Windows**:
    - Scalable but limited compared to Linux

Linux is used in **cloud systems & data centers**

2.6 Compatibility

- **Linux**:
    - Best for:
        - PHP
        - MySQL
    - Limited support for .NET
- **Windows**:
    - Best for:
        - .NET
        - Visual Basic

Choose based on **software requirements**

2.7 Security

- **Linux**:
    - More secure
    - Strong permission system
    - Less vulnerable to viruses
- **Windows**:
    - More prone to malware
    - Needs antivirus

Linux is preferred for **secure systems**

2.8 File System

- **Linux**:
  - Uses ext2, ext3, ext4
- **Windows**:
  - Uses NTFS, FAT

Different file systems → compatibility differences

2.9 Software & Support

- **Linux**:
  - Free software available
  - Community support
- **Windows**:
  - Paid software
  - Official Microsoft support

2.10 Use Cases

Linux is used in:

- Servers
- Cloud computing
- Embedded systems
- Cybersecurity

Windows is used in:

- Personal computers
- Office environments
- Gaming

2.11 Advantages & Disadvantages

Linux Advantages:

- Free and open-source
- Stable and secure
- Highly customizable

Linux Disadvantages:

- Steeper learning curve
- Limited commercial software

Windows Advantages:

- Easy to use
- Wide software support
- Good GUI

Windows Disadvantages:

- Expensive
- Less secure
- Frequent updates

2.12 Conclusion

- Linux is best for **performance, stability, and servers**
- Windows is best for **ease of use and general applications**

Final choice depends on:

- Application needs
- Budget
- Technical expertise